Monica Jin

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EDUCATION

North Carolina State University Expected May 2027

B.S. Computer Science, Minor in Art & Design | GPA: 4.0 *Raleigh, NC*

Honors: Park Scholar • National Merit Scholar • Salesforce Tech Accelerator (1 of 100 students selected nationwide)

EXPERIENCE

Salesforce May 2026 – Aug 2026

Software Engineer Intern, Event Monitoring & Security *San Francisco, CA*

- Shipped user-agent telemetry across **11 event log types** so enterprise customers can identify AI agents and connected apps accessing their data, leading delivery end to end across **8+ engineering teams**.
- Cut customer log-export bandwidth by **90–95%** for enterprise-scale log retrieval by building server-side compression in Java.
- Enabled customers to detect potential data exfiltration by designing and shipping security audit logs and APIs (Java, Apex) that flag suspicious GraphQL query activity.

IRIS Technologies Aug 2025 – May 2026

Software Engineer Intern *Raleigh, NC*

- Shipped v1.0 of a voice-first humanoid robot assistant (Unitree G1) for student-housing properties, built to run at events **without engineering supervision**, by owning the full software stack from design to on-site deployment.
- Delivered real-time conversational AI on the robot by integrating Claude API dialogue, speech recognition and synthesis, an iPad operator console, and LED state feedback (Python, C++), validating motion in NVIDIA Isaac Sim before hardware.

Pearson Jun 2025 – Aug 2025

Software Engineer Intern, AI/ML Platform *Durham, NC*

- Improved the quality of document chunks used for retrieval (RAG) by **80%** by building and deploying a Python image-processing pipeline that describes images with GPT-4.1 and filters unsafe content with Azure Content Safety.
- Launched video processing as a new LLM platform capability, generating transcripts and visual summaries with Whisper, GPT-4.1, and MoviePy.

PROJECTS

Ceramic AI: Machine Learning Glaze Predictor | *Python, scikit-learn, React, Three.js, TypeScript* Live demo | GitHub

- Shipped a **live web app** that helps ceramicists narrow down glaze recipes before studio testing, previewing predicted color, transparency, and surface on a **real-time 3D model** (Three.js).
- Trained scikit-learn models reaching **67% accuracy across 10 material classes** (2.8× the majority-class baseline) on 374 glazes; funded by an **NC State Arts + Science grant** and featured on the university's AI podcast.

Custom WebGL Rendering Engine | *WebGL, GLSL, JavaScript* GitHub

- Wrote a **3D rendering engine from scratch** with no graphics libraries: custom GLSL lighting shaders, **multi-pass rendering** for a live minimap, terrain, and real-time scene updates (rebuild of the arcade game Battlezone).

RESEARCH

Machine Learning Research, NC State | *Computer Vision, Adversarial Robustness* Jan 2024 – May 2025

- Exposed a **38.3% accuracy drop** in self-driving vision transformers across 10,000+ adversarial images, then built physical adversarial patches to test the attacks under real-world camera conditions.

Graphics & HCI Research, NC State | *Interface Latency* Aug 2025 – May 2026

- Helped run user studies on how people perceive **interface latency** in menu navigation and handwriting input, and tested whether rendering approach changes visible screen tearing.

LEADERSHIP & TEACHING

Teaching Assistant, CSC 201 (Applied AI for Engineering): built AI lab curriculum for **1,500+ students/year**.
Communications Officer, NC State VGDC • Adobe Student Ambassador • SIGGRAPH Volunteer • STEM Volunteer, Rwanda

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, C++, SQL, GLSL, Kotlin, C#, Apex
Web & Graphics: React, Three.js, WebGL, REST APIs, GraphQL, Figma, Blender
AI/ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, LLMs (GPT-4.1, Claude API), Whisper, RAG
Infrastructure: Git, Docker, Linux, Azure, CI/CD, Jenkins, Bazel, JUnit, NVIDIA Isaac Sim